



Surgical or Endovascular Management of Middle Cerebral Artery Aneurysms: A Randomized Comparison

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■ **OBJECTIVE:** There are few randomized data comparing clipping and coiling for middle cerebral artery (MCA) aneurysms. We analyzed results from patients with MCA aneurysms enrolled in the CURES (Collaborative UnRuptured Endovascular vs. Surgery) and ISAT-2 (International Subarachnoid Aneurysm Trial II) randomized trials.

■ **METHODS:** Both trials are investigator-led parallel-group 1:1 randomized studies. CURES includes patients with 3-mm to 25-mm unruptured intracranial aneurysms (UIAs), and ISAT-2 includes patients with ruptured aneurysms (RA) for whom uncertainty remains after ISAT. The primary outcome measure of CURES is treatment failure: 1) failure to treat the aneurysm, 2) intracranial hemorrhage during follow-up, or 3) residual aneurysm at 1 year. The primary outcome of ISAT-2 is death or dependency

(modified Rankin Scale score >2) at 1 year. One-year angiographic outcomes are systematically recorded.

■ **RESULTS:** There were 100 unruptured and 71 ruptured MCA aneurysms. In CURES, 90 patients with UIA have been treated and 10 await treatment. Surgical and endovascular management of unruptured MCA aneurysms led to treatment failure in 3/42 (7%; 95% confidence interval [CI], 0.02–0.19) for clipping and 13/48 (27%; 95% CI, 0.17–0.41) for coiling ($P = 0.025$). All 71 patients with RA have been treated. In ISAT-2, patients with ruptured MCA aneurysms managed surgically had died or were dependent (modified Rankin Scale score >2) in 7/38 (18%; 95% CI, 0.09–0.33) cases, and 8/33 (24%; 95% CI, 0.13–0.41) for endovascular. One-year imaging results were available in 80 patients with UIA and 62 with RA. Complete aneurysm occlusion

Key words

- Care trial
- Endovascular coiling
- Intracranial aneurysm
- MCA aneurysm
- Randomized trial
- Surgical clipping

Abbreviations and Acronyms

BRAT: Barrow Ruptured Aneurysm Trial
CI: Confidence interval
CT: Computed tomography
CURES: Collaborative UnRuptured Endovascular Versus Surgery
ISAT: International Subarachnoid Aneurysm Trial
ISFD: Intracranial flow disruptor
MCA: Middle cerebral artery
MRA: Magnetic resonance angiography
mRS: Modified Rankin Scale
RA: Ruptured aneurysm
RCT: Randomized controlled trial
SAH: Subarachnoid hemorrhage
UIA: Unruptured intracranial aneurysm

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was found in 30/40 (75%; 95% CI, 0.60–0.86) patients with UIA allocated clipping, and 14/40 (35%; 95% CI, 0.22–0.50) patients with UIA allocated coiling. Complete aneurysm occlusion was found in 24/34 (71%; 95% CI, 0.54–0.83) patients with RA allocated clipping, and 15/28 (54%; 95% CI, 0.36–0.70) patients with RA allocated coiling.

■ **CONCLUSIONS:** Randomized data from 2 trials show that better efficacy may be obtained with surgical management of patients with MCA aneurysms.

INTRODUCTION

Aneurysms located on the middle cerebral artery (MCA) are considered by many surgeons to represent a distinct subgroup of intracranial aneurysms for which clipping may still be the best management option. Most MCA aneurysms are accessible, proximal control can readily be secured, and clip application can typically proceed without requiring the dissection of perforating arteries. In comparison, certain anatomic features of MCA aneurysms such as a wide neck, often including a branch artery origin, frequently render endovascular management more difficult.^{1,2} New endovascular devices were and continue to be introduced to address these anatomic difficulties, including stents,³ flow diverters,⁴ and intrasaccular flow disruptors (ISFDs).⁵ Thus, although most aneurysms are increasingly treated with endovascular methods, many patients with MCA aneurysms are still managed surgically, but convincing evidence of which management paradigm is best is lacking.

ISAT (International Subarachnoid Aneurysm Trial) has shown that patients with good-grade, small, anterior circulation ruptured aneurysms (RAs) have better 1-year clinical outcomes after being coiled,⁶ a finding supported by BRAT (Barrow Ruptured Aneurysm Trial).⁷ However, there are many reasons to suspect that those results do not apply to aneurysms located at the MCA bifurcation. Only 14% of aneurysms in ISAT were on the MCA, likely because lesions in this location were preferentially treated with surgery. Even after selection, the MCA subgroup results were similar for coiling and clipping (relative risk, 1.01 [0.71–1.45]).⁸ In the all-inclusive BRAT trial, 21/31 (68%) of MCA aneurysms assigned to coiling were crossed over to clipping.⁹

For unruptured aneurysms, comparative case series suggest that clipping of MCA aneurysms may lead to better long-term anatomic outcomes¹⁰ but all available data are subject to the shortcomings of observational studies. The use of novel endovascular devices such as flow diverters and ISFDs is increasingly reported for unruptured MCA aneurysms but they have been associated with substantial perioperative morbidity or suboptimal long-term angiographic results.^{4,5}

Although patients and clinicians are drawn toward novel endovascular options, surgical clipping is “fighting an uphill battle with fewer and fewer troops.”¹¹ More than expert opinion and case series may be required to justify open neurosurgical aneurysm repair in the future. There are ongoing comparative trials addressing the best treatment of unruptured intracranial aneurysms (UIAs) and

RAs in all locations.^{12,13} These all-inclusive trials do not take into consideration the specific nature of MCA aneurysms and their treatment. Our goal is to present the MCA aneurysm results from CURES (Collaborative UnRuptured Endovascular vs. Surgery) and ISAT-2, not because they provide final answers to the clinical dilemma, but because they may serve to inform the conduct of future trials designed to offer optimal care to patients with MCA aneurysms.

METHODS

This report was prepared in the spirit of the CONSORT (Consolidated Standards of Reporting Trials) recommendations,¹⁴ but this pooled subgroup comparison is not a report of a stand-alone trial. The study protocols for both the CURES and the ISAT-2 trials (registered at Clinical trial.gov NCT01139892, NCT01668563) have been published, and interim results for each trial reported in 2017 and 2019, respectively.^{12,13,15,16} CURES and ISAT-2 are investigator-led, pragmatic randomized care trials conducted in North American and European centers. Data capture and management are through secure servers in compliance with Good Clinical Practice requirements. Protocols are approved by local institutional review boards of each participating institution and all patients (or representatives) provided written informed consent. Case report forms are simple, and required data collection made sparingly to facilitate completion by normal care personnel. Screening logs are not required per protocol.

Patients

All patients with MCA aneurysms included in CURES from inception in September 2010 to July 2020 and all patients with MCA aneurysms included in ISAT-2 from inception in November 2012 to July 2020 are reported. CURES includes independent (modified Rankin Scale [mRS] score <3), adult patients with an intradural unruptured aneurysm measuring 3–25 mm, with life expectancy >10 years. ISAT-2 includes any adult patient with an intradural aneurysm that has ruptured within the last 30 days, for whom the results of the initial ISAT⁶ are judged to not necessarily apply. Patients with grade 5 subarachnoid hemorrhage (SAH) are excluded. The patients with MCA aneurysms in CURES and ISAT-2 were enrolled at 3 Canadian, 1 American, and 3 European participating centers (Montréal, Edmonton, Ottawa, New York, Lille, Liège, and Barcelona).

Interventions

Patients are treated with surgical clipping or endovascular techniques as per local practices, with technical details left to individual operators. The use of stents, flow diverters, and other novel endovascular devices or adjunctive methods is recorded.

Outcome Measures and Blinding

The primary composite outcome measure of CURES is treatment failure, defined as 1) failure of intended modality to treat the aneurysm, 2) intracranial hemorrhage during follow-up, or 3) a residual aneurysm at 1 year. A crossover from one treatment modality to the other with no attempt to treat the aneurysm was not considered to reach the primary outcome of treatment failure. CURES also records clinical outcomes (mRS score) at discharge

and 1 year, by treating physicians not blinded to treatment assignment (as part of routine care).

The primary outcome of ISAT-2 is death or dependency (mRS score ≥ 2) at 1 year. A secondary outcome, imaging results (using catheter, computed tomography [CT], or magnetic resonance angiography [MRA]) at 1 year was assessed according to a pre-defined scale¹⁷: complete occlusion, residual neck, or residual aneurysm. Blinding to treatment assignment of patients, physicians, and outcome assessors was not possible.

Randomization

For both studies, parallel-group randomization is 1:1, concealed, and generated by a secure platform (MedSciNet) available 24/7. Details of minimization algorithms are available.^{15,16} In May 2016, the protocols were modified at 1 center (Centre Hospitalier de l'Université de Montréal, Montréal) to enable prerandomization.¹⁸

Statistical Analyses

Interim subgroup results from ongoing randomized trials are not usually reported.¹⁹ We restricted the use of statistical tests to a Fisher's exact test of the primary outcome of patients with MCA aneurysms recruited in CURES. Comparisons of baseline characteristics were by visual inspection only. All reported proportions are provided with 95% confidence intervals, using the Wilson score method to allow for small numbers.

Role of the Funding Source and Data Integrity

CURES was funded by the Canadian Institutes of Health Research (MOP 119554); ISAT-2 has not received any funding. Both studies are sponsored by Centre Hospitalier de l'Université de Montréal, Montréal. Neither funding agency nor sponsor had any part in study design, data collection, analysis, or reporting. The full study protocols have been published.^{15,16}

RESULTS

Unruptured MCA Aneurysms

There were 100 patients with unruptured MCA aneurysms, or 30% of the 330 patients enrolled in CURES to date. Patient and aneurysm characteristics for the surgical and endovascular groups are presented in [Table 1](#), and the patient flowchart in [Figure 1](#). Data presented in [Tables 2](#) and [3](#) are intent-to-treat unless otherwise specified; results of the as-treated analysis are available in [Supplementary Table 1](#) (online).

Ten patients with UIA have been enrolled but have not yet been treated (9 surgical and 1 endovascular).

There were 5/90 (6%) crossovers. Three patients with MCA UIA crossed over from endovascular to surgical management (all patient choice), and 2 crossed over from surgical to endovascular management (also patient choice). One surgically managed patient with an UIA (1/42; 2%) required an extracranial-intracranial bypass. Endovascular UIA treatment consisted of simple coiling $\pm$ balloon assistance in 44/48 (92%) of patients, with adjunctive devices infrequently used (stents, $n = 7$; flow diverter, $n = 1$; or ISFD, $n = 1$).

Length of hospitalization after clipping was longer than after endovascular treatment (median of 5.0 days after surgery and 1.0 day for endovascular, whereas new neurologic deficits after

treatments were similar (6/42 (14%; 95% confidence interval [CI], 0.07–0.28)) for surgery and 8/48 (17%; 95% CI, 0.09–0.30) for endovascular, including 1 intraprocedural death) ([Table 3](#)).

The number of surgically managed patients with unruptured MCA aneurysms who experienced the primary end point of treatment failure at 1 year is 3/42 (7%; 95% CI, 0.02–0.19) compared with 13/48 (27%; 95% CI, 0.17–0.41) for endovascular ($P = 0.025$; Fisher's exact test) ([Table 2](#)). Safety results (discharge mRS score) are available for all treated patients, and 1-year mRS score results are available for 80/90 (89%). At 1 year, surgical management of unruptured MCA aneurysms led to 0/42 (0%; 95% CI, 0–0.08) and endovascular management to 1/48 (2.1%; 95% CI, 0–0.11) dead or dependent (mRS score ≥ 2) patients ([Table 3](#)).

One-year imaging results were available in 80/90 (89%) treated patients with UIA ([Table 3](#)). For surgically managed patients, most of the 1-year imaging determinations (34/40; 85%) were on the basis of CT angiography. Endovascularly managed patients had most of the 1-year imaging determinations made on the basis of MRA (26/40; 65%). Complete MCA aneurysm occlusion was found in 30/40 patients allocated clipping (75%; 95% CI, 0.60–0.86) and 14/40 patients with UIA allocated coiling (35%; 95% CI, 0.22–0.50). Details regarding treatment-related complications and clinical deteriorations during follow-up are presented in [Supplementary Table 2](#).

None of the patients with surgically managed UIA had the index aneurysm retreated during follow-up, whereas 3 patients allocated endovascular management required retreatment. An example of MCA aneurysm recurrence in CURES that required retreatment is presented in [Figure 2](#).

Ruptured MCA Aneurysms

Seventy-one of the 185 patients (38%) enrolled in ISAT-2 had ruptured MCA aneurysms. Patient and aneurysm characteristics for the surgical and endovascular groups are presented in [Table 1](#), and the flowchart in [Figure 1](#).

There were 5 crossover patients (7%) in ISAT-2: 2 patients with RA crossed over from endovascular to surgical management (1 patient choice and 1 crossover by the treatment team), and 3 from surgical to endovascular management (2 patient choice and 1 by the treatment team). Intraparenchymal hematomas > 2 cm were present before inclusion in 9/38 surgical patients (24%) and 7/33 endovascular patients (21%). One patient with RA was treated with stent-assisted coiling and one with an ISFD.

Regarding the primary clinical outcome at 1 year, ruptured patients with MCA aneurysms allocated surgical management were dead or dependent (mRS score ≥ 2) in 7/38 patients (18%; 95% CI, 0.09–0.33), compared with 8/33 patients (24%; 95% CI, 0.13–0.41) allocated endovascular management ([Table 2](#)).

One-year imaging results were available in 62/71 (87%) of patients with RA ([Table 3](#)). Four included patient images were from before the 1-year time point (last observation carried forward). For surgically managed patients, most of the 1-year imaging determinations (31/34; 91%) were on the basis of CT angiography. Endovascularly managed patients had 1-year imaging determinations made on the basis of MRA (13/28; 46%) or digital subtraction angiography (12/28; 43%). Eight patients had no imaging because they had died. Complete MCA aneurysm occlusion

Table 1. Characteristics of Treated Patients with Middle Cerebral Artery Aneurysms from CURES and ISAT-2

Characteristics	CURES		ISAT-2	
	Allocated Surgery (n = 42)	Allocated Endovascular (n = 48)	Allocated Surgery (n = 38)	Allocated Endovascular (n = 33)
Patient				
Female sex, n (%)	32 (76)	36 (75)	25 (66)	25 (76)
Age at treatment, mean, years (SD)	57 (10)	60 (9)	58 (14.3)	55.4 (10.3)
Number of patients <50 years old (%)	11 (26)	6 (13)	10 (26)	8 (24)
Number of patients ≥70 years old (%)	1 (2)	5 (10)	6 (16)	2 (6)
Pretreatment (UIA)/prerupture (RA) mRS score = 0 n, (%)	33 (79)	37 (77)	32 (84)	32 (97)
Pretreatment (UIA)/prerupture (RA) mRS score = 1 n, (%)	8 (19)	10 (21)	5 (13)	0
Pretreatment (UIA)/prerupture (RA) mRS score = 2 n, (%)	1 (2)	1 (2)	1 (3)	1 (3)
Index aneurysm				
Left (%)	22 (52)	19 (40)	13 (34)	17 (52)
Size (mm), mean (SD), median, range	7.6 (3.2), 7, 3–18	8.0 (3.0), 8, 3–16	8.4 (6.4), 6, 2–29	7.8 (4.5), 6, 3–20
≤3 (%)	1 (2)	1 (2)	7 (18)	3 (9)
4–9 (%)	33 (79)	35 (73)	20 (53)	21 (64)
≥10 (%)	8 (19)	12 (25)	11 (29)	9 (27)
Wide-necked (≥4 mm) (%)	12 (29)	19 (40)	13 (34)	15 (45)
Recurrent, previously treated index aneurysm	1 (2)	0	0	0
Unruptured aneurysms				
Symptomatic aneurysm	11 (26)	3 (6)	—	—
Pain/headache	6 (14)	2 (4)	—	—
Stroke/transient ischemic attack	3 (7)	1 (2)	—	—
History of subarachnoid hemorrhage from another aneurysm (UIAs)	4 (10)	6 (13)	—	—
Hypertension	24 (57)	23 (48)	—	—
Current smoker	18 (43)	24 (50)	—	—
Excessive alcohol	6 (14)	2 (4)	—	—
Positive family history	9 (21)	9 (19)	—	—
Ruptured aneurysms				
WFNS grade 1	—	—	19 (50)	14 (42)
WFNS grade 2	—	—	10 (26)	13 (39)
WFNS grade 3	—	—	3 (8)	4 (12)
WFNS grade 4	—	—	6 (16)	2 (6)
Intraparenchymal hematoma >2 cm	—	—	9 (24)	7 (21)
Pretreatment infarction	—	—	2 (5)	2 (6)
Fisher grade 0–2	—	—	18 (47)	11 (33)
Grade 3–4	—	—	20 (53)	22 (67)
Treatment				
Time from randomization to treatment (UIAs, weeks; RAs, days) mean (SD)	9.6 (10.1)	8.3 (12.5)	0.9 (2.1)	0.5 (0.6)
Adherence to randomly allocated treatment	40 (95)	45 (94)	35 (92)	31 (94)

Continues

Table 1. Continued

Characteristics	CURES		ISAT-2	
	Allocated Surgery (n = 42)	Allocated Endovascular (n = 48)	Allocated Surgery (n = 38)	Allocated Endovascular (n = 33)
Aneurysm (re)rupture after randomization, before treatment	0	0	0	1 (3)
Extracranial-intracranial bypass	1 (2)	0	0	0
Stent-assisted coiling	1* (2)	7 (15)	0	1 (3)
Flow diverter-assisted coiling	0	1 (2)	0	0
Intrasaccular flow disruptor (Woven EndoBridge)	0	1 (2)	0	1 (3)

SD, standard deviation; UIA, unruptured intracranial aneurysm; World Federation of Neurosurgical Societies; RA, ruptured aneurysm; mRS, modified Rankin Scale.
*Crossed over from clipping to coiling (patient choice).

was found in 24/34 patients allocated clipping (71%; 95% CI, 0.54–0.83) and 15/28 patients allocated coiling (54%; 95% CI, 0.36–0.70). Details regarding new neurologic deficits, aneurysm ruptures during follow-up, and additional treatments directed

against the index aneurysm are presented in [Supplementary Table 2](#).

No surgically managed patients with RA required retreatment of the index aneurysm during follow-up, whereas 4 patients who

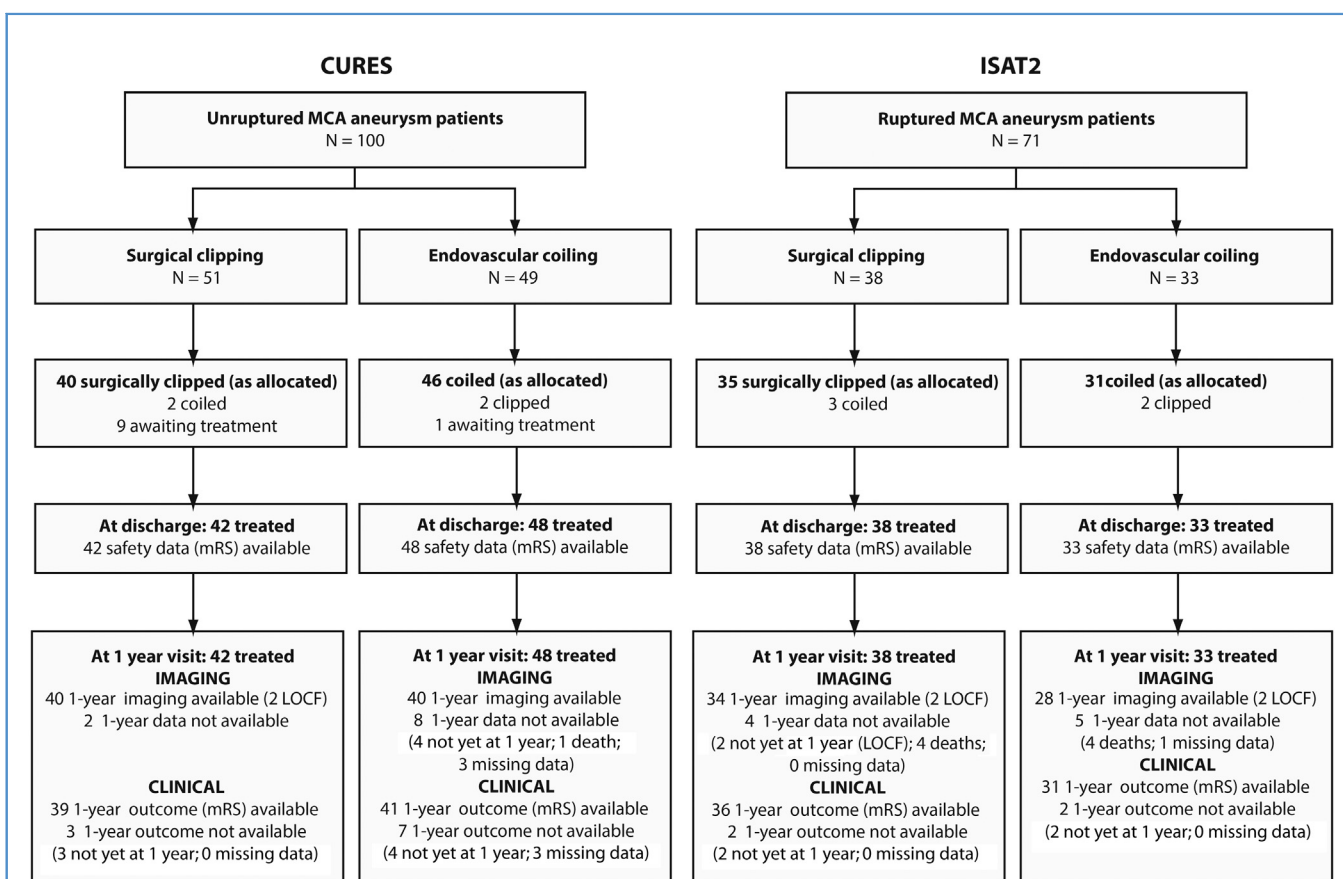


Figure 1. Flowcharts of the CURES trial and the ISAT-2 studies. LOCF, last observation carried forward; MCA, middle cerebral artery; mRS, modified Rankin Scale.

Table 2. Primary End Points Reached in CURES and ISAT-2 for Patients with Middle Cerebral Artery Aneurysms

Treatment Failure	Surgery	Endovascular	P
Intent-to-treat analysis of primary trial end point			
CURES: unruptured MCA aneurysms			
Failure of modality to treat aneurysm	0	4*	
Hemorrhage during follow-up	0	0	
Residual aneurysm at 1 year	3	9	
Total	3/42 (7%) (95% CI, 0.02–0.19)	13/48 (27%) (95% CI, 0.17–0.41)	0.025
ISAT-2: ruptured MCA aneurysms			
mRS score >2 at 1 year	7/38 (18%) (95% CI, 0.09–0.33)	8/33 (24%) (95% CI, 0.13–0.41)	—
As-treated analysis of primary trial end point			
CURES: unruptured MCA aneurysms			
Failure of modality to treat aneurysm	0	4*	
Hemorrhage during follow-up	0	0	
Residual aneurysm at 1 year	2	10	
Total	2/43 (5%) (95% CI, 0.01–0.15)	14/47 (30%) (95% CI, 0.19–0.44)	0.002
ISAT-2: Ruptured MCA aneurysms			
mRS score >2 at 1 year	7/37 (19%) (95% CI, 0.09–0.34)	8/34 (24%) (95% CI, 0.12–0.40)	—

MCA, middle cerebral artery; CI, confidence interval; mRS, modified Rankin Scale.
 *3 untreated (fully patent) aneurysms after procedure, 1 intraprocedural death

received endovascular management required retreatment (Table 3). Two ISAT-2 patients treated with coiling experienced rebleeding from the index MCA aneurysm during follow-up. One patient allocated coiling with a partially thrombosed 20-mm right MCA aneurysm rebled 3 months after coiling, which was treated by repeat coiling. The aneurysm was found to recur again angiographically within 3 months, and the patient underwent surgical treatment. The second patient had an 8-mm right MCA aneurysm and was allocated surgery but chose to cross over to endovascular treatment. Coiling led to complete initial aneurysm occlusion, and she recovered to an mRS score of 1 by 6 months after SAH. At 9 months after treatment, the patient had a fatal repeat SAH from the coiled recurrent aneurysm.

DISCUSSION

Patients with MCA aneurysms randomly allocated to surgical clipping had similar clinical outcomes (number of patients with mRS score >2) but better treatment efficacy outcomes at 1 year than did patients allocated endovascular treatment. These findings require confirmation in a future trial dedicated to patients with MCA aneurysms.

Our study differs from previous randomized evidence regarding ruptured MCA aneurysms. First, ISAT included disproportionately few selected patients with MCA aneurysms (301/2143 or 14%, compared with 38% in ISAT-2) recruited between 1994 and 2004, a time when only simple coiling was available. Although ISAT showed better clinical outcomes for coiled aneurysms at 1 year, this overall trial result was not confirmed for MCA aneurysms. The

clinical primary end point of mRS score >2 was reached in 39/139 clipped patients (28.1%; 95% CI, 0.21–0.36) and 46/162 coiled patients (28.4%; 95% CI, 0.22–0.36).⁸ However, subgroup findings that differ from overall trial results should always be interpreted with caution.²⁰

The suspicion that only selected patients with MCA aneurysms were judged eligible for endovascular treatment at the time of ISAT is supported by the prerandomized BRAT study: of 61 patients with ruptured MCA aneurysms included between 2003 and 2007, 30 were assigned clipping and 31 coiling. Of the 31 (68%) endovascular patients, 21 were crossed over to the surgical arm. Thus, it was reported that “direct comparison of the treatments was impossible.”⁹ The Finnish randomized controlled trial (RCT) on clipping versus coiling^{21,22} reported only 19 ruptured MCA aneurysms because 59 patients with MCA aneurysms were excluded. One RCT reported in 2012²³ included only 35 patients with ruptured MCA aneurysms and it did not provide angiographic results per aneurysm location. By contrast, there were few RA crossovers (7%) in the present study. If most MCA aneurysms can be treated endovascularly, clinical results of contemporary technical achievements remain to be properly compared with surgical clipping. In particular, although overall clinical results were similar at 1 year in ISAT-2, rebleedings and retreatments after endovascular treatment are worrisome. This randomized comparison of angiographic results for ruptured MCA aneurysms is the largest to date because angiographic follow-ups were not required in ISAT.

ISAT-2 was designed to be, and can still be considered, a continuation of the original ISAT trial, with a superiority

Table 3. Clinical Outcome (Intent-to-Treat) and Angiographic Outcomes of Treated Patients

Clinical Outcomes	CURES		ISAT2	
	Surgical (n = 42)	Endovascular (n = 48)	Surgical (n = 38)	Endovascular (n = 33)
Number of days hospitalized peritreatment, median (interquartile range)	5.0 (4.0–7.0)	1.0 (1.0–2.0)	15.5 (12.0–24.0)	12.0 (9.0–18.5)
Number of patients hospitalized for >5 days (unruptured intracranial aneurysm), >20 days (ruptured aneurysm)	20 (48)	4 (8)	14 (37)	6 (18)
Patients with new neurologic deficits after treatment	6 (14)	8 (17)	2 (5)	7 (21)
Discharge location				
Home	39 (93)	46 (96)	23 (61)	18 (55)
Other hospital or rehabilitation center	3 (7)	1 (2)	13 (34)	12 (36)
Death	0	1 (2)	2 (5)	3 (9)
mRS score at discharge				
0	24 (57)	37 (77)	7 (18)	11 (33)
1	12 (29)	6 (13)	14 (37)	7 (21)
2	6 (14)	3 (6)	8 (21)	4 (12)
3	0	1 (2)	6 (16)	7 (21)
4	0	0	1 (3)	0
5	0	0	0	1 (3)
6	0	1 (2)	2 (5)	3 (9)
Total	42	48	38	33
mRS score at 1 year				
1	16 (38)	6 (13)	13 (34)	8 (24)
2	0	4 (8)	5 (13)	3 (9)
3	0	0	3 (8)	3 (9)
4	0	0	0	0
5	0	0	0	1 (3)
6	0	1 (2)	4 (11)	4 (12)
Not yet at 1 year follow-up/missing data	3 (7)	7 (15)	2 (5)	2 (6)
Total	42	48	38	33
Additional procedures during hospitalization for subarachnoid hemorrhage				
Directed against index ruptured middle cerebral artery aneurysm	—	—	0	3 (9)
Decompressive craniectomy	—	—	3 (8)	2 (6)
Ventriculoperitoneal shunt/permanent cerebrospinal fluid diverting procedure	—	—	5 (13)	0
Tracheostomy and/or percutaneous endoscopic gastrostomy	—	—	3 (8)	2 (6)
Symptomatic vasospasm/requiring balloon angioplasty	—	—	10 (26)/1 (3)	6 (18)/1 (3)
After discharge				
Rerupture of index aneurysm during follow-up	0	0	1* (3)	1 (3)
Retreatment of index aneurysm during follow-up	0	3 (6)	0	4 (12)
mRS, modified Rankin Scale. *Crossed over from clipping to coiling (patient choice). †Patent aneurysms after attempted coiling.				

Continues

Table 3. Continued

Angiographic Outcomes, 1 year, Intent-to-Treat	Surgical (n = 40)	Endovascular (n = 40)	Surgical (n = 34)	Endovascular (n = 28)
Complete occlusion	30 (75)	14 (35)	24 (71)	15 (54)
Residual neck	7 (18)	14 (35)	6 (18)	8 (29)
Residual aneurysm	3 (8)	9 + 3† (30)	4 (12)	5 (18)
Total	40	40	34	28
Angiographic Outcomes, 1 year, As-Treated	Surgical (n = 41)	Endovascular (n = 39)	Surgical (n = 34)	Endovascular (n = 28)
Complete occlusion	30 (73)	14 (36)	26 (76)	13 (46)
Residual neck	9 (22)	12 (31)	5 (15)	9 (32)
Residual aneurysm	2 (5)	10 + 3† (33)	3 (9)	6 (21)
Total	41	39	34	28

mRS, modified Rankin Scale.
 *Crossed over from clipping to coiling (patient choice).
 †Patent aneurysms after attempted coiling.

hypothesis in favor of endovascular treatment.¹⁶ The results presented here suggest that this hypothesis may not be appropriate for ruptured MCA aneurysms. Showing a result for a subgroup (the MCAs) that differs from the overall results at the end of ISAT-2 risks being scientifically problematic just as in the original ISAT. For this reason, a new trial dedicated to ruptured MCA aneurysms is needed.

With respect to unruptured MCA aneurysms, other than from CURES,¹² there is no randomized evidence. An abundance of observational data exists from multiple case series.^{24–26} The

most recent meta-analyses of case series of unruptured MCA aneurysms have shown superiority of clipping in terms of safety in one report, and superiority of coiling in another, but numbers remain small and uncertain.^{24,26} These meta-analyses reported complete occlusion rates of 95% (95% CI, 0.90–0.98) with clipping compared with 60% (95% CI, 0.45–0.74) with coiling.²⁶ The high rates of complete occlusions in reported case series compared with the randomized evidence presented here are likely related to publication bias, although a recent nonrandomized comparison of clip-first and coil-first paradigms

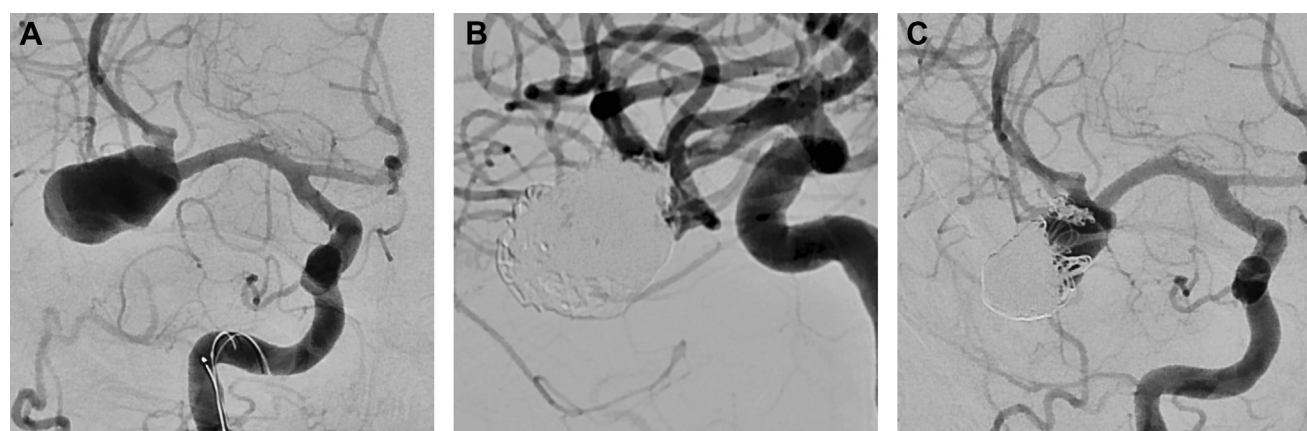


Figure 2. Angiographic images showing an unruptured 18-mm right middle cerebral artery aneurysm in a 77-year-old woman before elective coiling (**A**). Complete aneurysm occlusion was obtained at the completion of the

endovascular procedure (**B**; projection chosen to show aneurysm neck). At 1 year follow-up (**C**), a major recurrent aneurysm was found.

for MCA aneurysms reported complete occlusion rates of 86% (69/80) for clipped aneurysms versus 27% (20/74) for coiled aneurysms at 6–12 months follow-up.²⁷

Meta-analyses of the angiographic results of coiling of MCA aneurysms focus on “complete or near-complete occlusion” or “adequate occlusion” rates.^{28,29} The choice to assimilate complete occlusions with aneurysm necks comes from RA studies, because both results have been shown to prevent rebleeding. Complete occlusions may be more pertinent to the management of UIAs, because incomplete occlusions and long-term recurrences still affect many endovascular options while long-term protection from rupture remains the goal of therapy.

All available evidence on unruptured MCA aneurysms is observational and all is affected by systemic biases. The present study is the only randomized comparison of endovascular and surgical treatment of unruptured MCA aneurysms and results are compatible with a hypothesis in favor of clipping, because treatment failures were more common after endovascular treatment.

The current study brings the issue of the role of trials in clinical care to the fore. At the very least, to justify making treatment decisions in practice, we need to ensure that the alleged increased efficacy of any more invasive, potentially more risky or novel option has been proven in a rigorous RCT, a task that remains to be accomplished for surgical clipping as well as for most recent endovascular innovations.^{30–33} Some novel endovascular options may claim that they can lead to complete aneurysm occlusions more often than historical endovascular controls, but they have never been properly compared with the results of surgical clipping.⁵

Patients with MCA aneurysms were frequently recruited in both CURES and ISAT-2, a sign of the ongoing clinical uncertainty regarding the proper management of these patients. CURES and ISAT-2 are care trials designed to guide practice in the presence of uncertainty.³⁴ However, these trials have not been popular with clinicians and neurovascular centers and recruitment has been slow, perhaps as a consequence of their pragmatic design and all-inclusive nature. Patients with MCA aneurysms may need a specific trial to find out what is the best management of their condition. For clinicians, a benchmark level of surgical results obtained in patients with MCA aneurysms whose treatments have been randomly allocated is needed to establish the merits of clipping and serve as a reference for which other trials could aim.

The problem is how to determine what constitutes optimal care, while simultaneously trying to provide optimal care. The solution is an RCT designed in the best interests of participating patients.³⁵ Clinicians can then offer surgical clipping as optimal care in the presence of uncertainty through 1:1 randomized allocation of surgery versus any endovascular option. The additional merit of such a trial would be that it could help maintain the open neurosurgical skills necessary for safe neurosurgical management in a field now threatened by obsolescence,¹¹ at the same time offering an opportunity to participate in the progress of endovascular therapy. The broad outlines of a potential new trial dedicated to MCA aneurysms is presented in [Supplementary Table 3](#).

Many surgeons believe clipping to be superior in terms of long-term angiographic results and it would be tempting to design the new study accordingly. However, there are multiple

problems with using a surrogate angiographic outcome measure.³⁶ One systematic difference between the 2 treatments concerns the use of follow-up imaging studies (CT angiography for clipped and MRA for coiled MCA aneurysms). The diagnostic accuracy of each method in detecting aneurysm residua differs.^{37,38} Requiring 1-year catheter angiography for all patients is unlikely to be clinically acceptable and is contrary to the pragmatic care trial philosophy.³⁴ Moreover, it is uncertain whether “same-sized” postcoiling and postclipping residua, even on catheter angiography, have the same clinical significance. Our studies suggest more frequent complete occlusions with clipping. The crucial empirical question, for clipping as for any endovascular innovation, is how much more morbidity (if any) is justified to achieve better angiographic results, and whether angiographic results are predictive of better long-term outcomes.

Limitations of this study include that it is an unpowered subgroup analysis of 2 ongoing trials; some patients have yet to be treated, and not all treated patients have 1-year follow-up results. The number of patients is small, and the imaging evaluations were not performed by a core laboratory. The choice to compare “complete aneurysm occlusion” rates is debatable. Thus the findings reported here qualify as exploratory. They should be interpreted with caution and be used only as hypotheses for future trials.³⁹ Only a few (10/48, 21%) of the endovascularly managed cases in this work used adjunctive devices such as stents, flow diverters, or ISFDs. The hypothesis that more liberal use of novel endovascular adjuncts could decrease the incidence of treatment failure or improve angiographic results should be tested in randomized trials.

We believe that our results should nevertheless encourage surgeons to participate in a new trial dedicated to aneurysms in this location. From the endovascular perspective, a variety of innovative minimally invasive solutions are increasingly tried for MCA aneurysms. These innovations should also be tested in a randomized trial.⁴⁰ Patients recruited in such trials should be offered at least a chance to be treated surgically. We suggest that the benchmark to be used to compare endovascular innovations for MCA aneurysms should be surgical clipping.

CONCLUSIONS

Randomized subgroup data from 2 trials show that better efficacy may be obtained with surgical management of patients with MCA aneurysm. This finding should be confirmed by randomized trials dedicated to MCA aneurysms.

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REFERENCES

- Iijima A, Piotin M, Mounayer C, Spelle L, Weill A, Moret J. Endovascular treatment with coils of 149 middle cerebral artery berry aneurysms. *Radiology*. 2005;237:611-619.
- Vanzin JR, Mounayer C, Piotin M, Spelle L, Boissonnet H, Moret J. [Endovascular treatment of unruptured middle cerebral artery aneurysms]. *J Neuroradiol*. 2005;32:97-108 [in French].
- Zhang X, Zuo Q, Tang H, et al. Stent assisted coiling versus non-stent assisted coiling for the management of ruptured intracranial aneurysms: a meta-analysis and systematic review. *J Neurointerv Surg*. 2019;11:489-496.
- Cagnazzo F, Mantilla D, Lefevre PH, Dargazanli C, Gascou G, Costalat V. Treatment of middle cerebral artery aneurysms with flow-diverter stents: a systematic review and meta-analysis. *AJNR Am J Neuroradiol*. 2017;38:2289-2294.
- Pierot L, Szikora I, Barreau X, et al. Aneurysm treatment with WEB in the cumulative population of two prospective, multicenter series: 3-year follow-up [e-pub ahead of print]. *J Neurointerv Surg*. <https://doi.org/10.1136/neurintsurg-2020-016151>, accessed June 12, 2020.
- Molyneux A, Kerr R, Stratton I, et al. International Subarachnoid Aneurysm Trial (ISAT) of neurosurgical clipping versus endovascular coiling in 2143 patients with ruptured intracranial aneurysms: a randomised trial. *Lancet*. 2002;360:1267-1274.
- McDougall CG, Spetzler RF, Zabramski JM, et al. The Barrow Ruptured Aneurysm Trial. *J Neurosurg*. 2012;116:135-144.
- Molyneux AJ, Kerr RS, Yu LM, et al. International subarachnoid aneurysm trial (ISAT) of neurosurgical clipping versus endovascular coiling in 2143 patients with ruptured intracranial aneurysms: a randomised comparison of effects on survival, dependency, seizures, rebleeding, subgroups, and aneurysm occlusion. *Lancet*. 2005;366:809-817.
- Mooney MA, Simon ED, Brigeman S, et al. Long-term results of middle cerebral artery aneurysm clipping in the Barrow Ruptured Aneurysm Trial. *J Neurosurg*. 2018;130:895-901.
- Smith TR, Cote DJ, Dasenbrock HH, et al. Comparison of the efficacy and safety of endovascular coiling versus microsurgical clipping for unruptured middle cerebral artery aneurysms: a systematic review and meta-analysis. *World Neurosurg*. 2015;84:942-953.
- Macdonald RL. Editorial: Clip or coil? Six years of follow-up in BRAT. *J Neurosurg*. 2015;123:605-607.
- Darsaut TE, Findlay JM, Magro E, et al. Surgical clipping or endovascular coiling for unruptured intracranial aneurysms: a pragmatic randomised trial. *J Neurol Neurosurg Psychiatry*. 2017;88:663-668.
- Darsaut TE, Roy D, Weill A, et al. A randomized trial of endovascular versus surgical management of ruptured intracranial aneurysms: interim results from ISAT2. *Neurochirurgie*. 2019;65:370-376.
- Zwarenstein M, Treweek S, Gagnier JJ, et al. Improving the reporting of pragmatic trials: an extension of the CONSORT statement. *BMJ*. 2008;337:a2390.
- Darsaut TE, Findlay JM, Raymond J. The design of the Canadian UnRuptured Endovascular versus Surgery (CURES) trial. *Can J Neurol Sci*. 2011;38:236-241.
- Darsaut TE, Jack AS, Kerr RS, Raymond J. International Subarachnoid Aneurysm Trial—ISAT part II: study protocol for a randomized controlled trial. *Trials*. 2013;14:156.
- Raymond J, Guilbert F, Weill A, et al. Long-term angiographic recurrences after selective endovascular treatment of aneurysms with detachable coils. *Stroke*. 2003;34:1398-1403.
- Raymond J, Darsaut TE, Roy DJ. Recruitment in clinical trials: the use of Zelen's prerandomization in recent neurovascular studies. *World Neurosurg*. 2017;98:403-410.
- Fleming TR, Sharples K, McCall J, Moore A, Rodgers A, Stewart R. Maintaining confidentiality of interim data to enhance trial integrity and credibility. *Clin Trials*. 2008;5:157-167.
- Rothwell PM. Treating individuals 2. Subgroup analysis in randomised controlled trials: importance, indications, and interpretation. *Lancet*. 2005;365:176-186.
- Koivisto T, Vanninen R, Hurskainen H, Saari T, Hernesniemi J, Vapalahti M. Outcomes of early endovascular versus surgical treatment of ruptured cerebral aneurysms. A prospective randomized study. *Stroke*. 2000;31:2369-2377.
- Vanninen R, Koivisto T, Saari T, Hernesniemi J, Vapalahti M. Ruptured intracranial aneurysms: acute endovascular treatment with electrolytically detachable coils—a prospective randomized study. *Radiology*. 1999;211:325-336.
- Li ZQ, Wang QH, Chen G, Quan Z. Outcomes of endovascular coiling versus surgical clipping in the treatment of ruptured intracranial aneurysms. *J Int Med Res*. 2012;40:2145-2151.
- Alreshidi M, Cote DJ, Dasenbrock HH, et al. Coiling versus microsurgical clipping in the treatment of unruptured middle cerebral artery aneurysms: a meta-analysis. *Neurosurg*. 2018;83:879-889.
- Mortimer AM, Bradley MD, Mews P, Molyneux AJ, Renowden SA. Endovascular treatment of 300 consecutive middle cerebral artery aneurysms: clinical and radiologic outcomes. *AJNR Am J Neuroradiol*. 2014;35:706-714.
- Tocaceli G, Diana F, Cagnazzo F, et al. Microsurgical clipping compared with new and most advanced endovascular techniques in the treatment of unruptured middle cerebral artery aneurysms: a meta-analysis in the modern era. *World Neurosurg*. 2020;137:451-464.e451.
- Berro DH, L'Allinec V, Pasco-Papon A, et al. Clip-first policy versus coil-first policy for the exclusion of middle cerebral artery aneurysms [e-pub ahead of print]. *J Neurosurg*. <https://doi.org/10.3171/2019.5.JNS19373>, accessed February 25, 2021.
- Brinjikji W, Lanzino G, Cloft HJ, Rabinstein A, Kallmes DF. Endovascular treatment of middle cerebral artery aneurysms: a systematic review and single-center series. *Neurosurg*. 2011;68:397-402 [discussion 402].
- Zijlstra IA, Verbaan D, Majoie CB, Vandertop P, van den Berg R. Coiling and clipping of middle cerebral artery aneurysms: a systematic review on clinical and imaging outcome. *J Neurointerv Surg*. 2016;8:24-29.

30. Darsaut TE, Raymond J. The design of the STenting in Aneurysm Treatments (STAT) trial. *J Neurointerv Surg.* 2012;4:178-181.
31. Kotowski M, Naggara O, Darsaut TE, et al. Safety and occlusion rates of surgical treatment of unruptured intracranial aneurysms: a systematic review and meta-analysis of the literature from 1990 to 2011. *J Neurol Neurosurg Psychiatry.* 2013;84:42-48.
32. Raymond J, Gentric JC, Darsaut TE, et al. Flow diversion in the treatment of aneurysms: a randomized care trial and registry. *J Neurosurg.* 2017;127:454-462.
33. Raymond J, Januel AC, Iancu D, et al. The RISE trial: a Randomized Trial on Intra-Saccular Endobridge devices. *Interven Neuroradiol.* 2020;26:61-67.
34. Raymond J, Darsaut TE, Altman DG. Pragmatic trials used as optimal medical care: principles and methods of care trials. *J Clin Epidemiol.* 2014;67:1150-1156.
35. Raymond J, Darsaut TE, Roy D. Care and research concepts should be revised to practice outcome-based medical care. *J Clin Epidemiol.* 2019;116:155-160.
36. Darsaut TE, Chapot R, Raymond J. Changing the rules of the game: the problem of surrogate angiographic outcomes in the evaluation of aneurysm treatments. *AJNR Am J Neuroradiol.* 2020;41:2174-2175.
37. Ahmed SU, Mocco J, Zhang X, et al. MRA versus DSA for the follow-up imaging of intracranial aneurysms treated using endovascular techniques: a meta-analysis. *J Neurointerv Surg.* 2019;11:1009-1014.
38. Uricchio M, Gupta S, Jakowenko N, et al. Computed tomography angiography versus digital subtraction angiography for postclipping aneurysm obliteration detection. *Stroke.* 2019;50:381-388.
39. Sackett D. The tactics of performing therapeutic trials. In: Haynes R, Sackett DR, Guyatt GH, Tugwell P, eds. *Clinical Epidemiology: How to Do Clinical Practice Research.* Philadelphia, PA: Lippincott, Williams & Wilkins; 2006:66-172.
40. Fahed R, Darsaut TE, Raymond J. The introduction of innovations in neurovascular care: patient selection and randomized allocation. *World Neurosurg.* 2018;118:e99-e104.

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SUPPLEMENTARY DATA

Supplementary Table 1. Clinical Outcome (as Treated)				
Clinical Outcomes	CURES		ISAT2	
	Surgical (n = 43)	Endovascular (n = 47)	Surgical (n = 37)	Endovascular (n = 34)
Number of days hospitalized peritreatment: median (interquartile range)	5.0 (4.0–7.0)	1.0 (1.0–2.0)	16.0 (12.5–24.0)	12.0 (9.0–15.3)
Number of patients hospitalized for: >5 days (unruptured intracranial aneurysm), >20 days (RA)	21 (49)	3 (6)	14 (38)	6 (18)
Patients with new neurologic deficits after treatment	7 (16)	7 (15)	2 (5)	7 (21)
Discharge location				
Home	39 (91)	46 (98)	22 (60)	19 (56)
Other hospital/rehabilitation center	4 (9)	0	13 (35)	12 (35)
Death	0	1 (2)	2 (5)	3 (9)
mRS score at discharge				
0	24 (56)	37 (79)	7 (19)	11 (32)
1	12 (28)	6 (13)	13 (35)	8 (24)
2	6 (14)	3 (6)	7 (19)	5 (15)
3	1 (2)	0	7 (19)	6 (18)
4	0	0	1 (3)	0
5	0	0	0	1 (3)
6	0	1 (2)	2 (5)	3 (9)
Total	43	47	37	34
mRS score at 1 year				
0	23 (53)	30 (64)	11 (30)	12 (35)
1	16 (37)	6 (13)	12 (32)	9 (26)
2	0	4 (9)	5 (14)	3 (9)
3	0	0	4 (11)	2 (6)
4	0	0	0	0
5	0	0	0	1 (3)
6	0	1 (2)	3 (8)	5 (15)
Not yet at 1 year follow-up/missing data	4 (9)	6 (13)	2 (5)	2 (6)
Total	43	47	37	34
Additional procedures during hospitalization for subarachnoid hemorrhage				
Directed against index ruptured middle cerebral artery aneurysm	—	—	1 (3)	2 (6)
Decompressive craniectomy	—	—	3 (8)	2 (6)
Ventriculoperitoneal shunt/permanent cerebrospinal fluid diverting procedure	—	—	5 (14)	0
Tracheostomy and/or percutaneous endoscopic gastrostomy	—	—	3 (8)	2 (6)
Symptomatic vasospasm/requiring balloon angioplasty	—	—	10 (27)/1 (3)	6 (18)/1 (3)
After discharge				
Rerupture of index aneurysm during follow-up	0	0	0	2 (6)
Retreatment of index aneurysm during follow-up	0	3 (6)	0	4 (12)
mRS, modified Rankin Scale.				

Supplementary Table 2. New Neurologic Deficits, Reruptures During Follow-Up, and Treatment Failures of Middle Cerebral Artery Aneurysms

New neurologic deficits	
CURES	
Surgery	Seizure after extracranial-intracranial bypass with thromboemboli into distal branches, leading to aphasia and hemiparesis
	Expressive aphasia after surgery
	Postoperative aphasia, resolved within 4 days
	(Crossover) Urgent unplanned stenting; branch occlusion treated with abciximab. Residual aphasia
	Not moving right arm postoperatively
	Receptive aphasia postoperatively
EVT	Branch occlusion, with distal emboli. Left temporal infarction with receptive aphasia
	Weakness of the right hand and aphasia after embolization. Resolved by time of discharge
	Death from intraprocedural subarachnoid hemorrhage
	Migrated coil, treated with urgent stenting. Left pronator drift after procedure
	Ischemic stroke with hemorrhagic transformation. Ongoing memory issues
	(Crossover) Aphasia with right sided weakness after surgery
	(Crossover) Postoperative seizures managed with antiepileptic medication
	New weakness of right leg after procedure
ISAT-2	
Surgery	Ischemic stroke after clipping; discharge modified Rankin Scale score 4, died 5 months later
	Arm numbness after aneurysm clipping
EVT	Delayed stroke 3 weeks after coiling. Ipsilateral to aneurysm; symptomatic with confusion and visual loss
	Stent thrombosis requiring urgent take-back to angiography suite. MCA stroke and death
	Stroke per-embolization. Died 4 days after coiling
	Asymptomatic infarction seen on postcoil computed tomography performed postprocedure day 1 (large enough to require decompressive craniectomy)
	Large intraparenchymal hematoma, contralateral to coiled aneurysm, but aneurysm had been treated with parent vessel occlusion. Death
	Embolitic shower on postprocedure day 2 causing aphasia, mild hemiparesis
Continues	

Supplementary Table 2. Continued

New neurologic deficits	
	Intraprocedural embolism leading to left MCA stroke
(Re)-ruptures during follow-up	
CURES	
Surgery	None
EVT	None
ISAT-2	
Surgery	(Crossover) Rebled from coiled aneurysm 9 months after treatment leading to death
EVT	Coiled, rebled from index aneurysm 3 months later, re-coiled. Recurrence growing again, clipped
Treatment failures	
CURES	
Surgery	None
EVT	Additional treatments required for index aneurysm (n = 3)
	Failed attempted coiling; sent for surgery 3 months later
	Failed attempted coiling; sent for surgery 1 year later
	Failed attempted coiling; sent for surgery 1 year later
	Intraprocedural subarachnoid hemorrhage (fatal) (1)
Additional treatments required for index aneurysm	
ISAT-2	
Surgery	None
EVT	Recurrent aneurysm found on follow-up. Retreated 6 months later, complicated by intraprocedural rupture requiring operative evacuation
	Coiled, rebled from index aneurysm 3 months later, re-coiled. Recurrence growing again, clipped
	Growing aneurysm recurrence seen on follow-up, surgically clipped
	Growing aneurysm recurrence identified during follow-up; retreated with coiling
EVT, endovascular treatment; MCA, middle cerebral artery.	

Supplementary Table 3. Outline of a New Trial Dedicated to Middle Cerebral Artery Aneurysms

Current Aneurysm Trials			
	CURES	ISAT-2	Future Trial on MCA Aneurysms
Trial design	Pragmatic randomized trial	Pragmatic randomized trial	Combined pragmatic randomized trials (2 trials)
Selection criteria	Adult UIAs 3–25 mm	Any non-ISAT aneurysm for which uncertainty remains	Any MCA aneurysm, ruptured or unruptured
Primary hypothesis	In favor of surgery	In favor of endovascular treatment	2 different hypotheses for ruptured aneurysms or UIAs (both in favor of surgery)
Primary outcome	Treatment failure (3 part composite)	1 year clinical outcome (mRS score >2)	Treatment failure (4 part composite)
	Failure to treat aneurysm		Failure to treat aneurysm
	Intracranial hemorrhage during follow-up		Intracranial hemorrhage during follow-up
	Residual aneurysm at 1 year		Residual aneurysm at 1 year
			mRS score >2 at 1 year
Randomization	1: 1 with minimization	1:1 with minimization	1: 1 stratified (ruptured vs. unruptured)
Number of patients	Approximately 330 unruptured aneurysms	Approximately 2000 ruptured aneurysms	Approximately 250 unruptured MCA aneurysms
			Approximately 400 ruptured MCA aneurysms
MCA, middle cerebral artery; UIA, unruptured intracranial aneurysm; mRS, modified Rankin Scale.			